

Complete MCP Server/Client Setup Guide.

MCP – Model Context Protocol

MCP is a guide for the AI's to choose the right API and interact with Third Party Platforms. MCP provides agents the context they need to interact with third party Platforms. Agents uses MCP Client to interact with MCP Servers.

MCP Clients are embedded within Coding Agents in IDE's like Cursor, Windsurf, Claude Code, Claude Desktop.

We could then get these agents to connect to a local Kubernetes cluster.

MCP Server Components: Resources, Tools, Prompts

MCP Client Components: Roots, Sampling, Elicitation.

Build an MCP Server

Let's build MCP Server - Scope of this MCP Server is to Restarts a pod by name in the web namespace on your Docker Desktop Kubernetes cluster.

This MCP Server Exposes two tools:

- `list_pods()` — lists pod names + status in the web namespace
- `restart_pod(pod_name)` — deletes the named pod (its controller, e.g. a Deployment, recreates it)

Install Docker Desktop:

Enable WSL 2 first (recommended backend):

```
#wsl -install
```

Restart when it finishes. Then confirm with:

```
#wsl -status
```

Install Docker Desktop:

```
#winget install -e --id Docker.DockerDesktop
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> wsl --status
Default Version: 2
WSL1 is not supported with your current machine configuration.
Please enable the "Windows Subsystem for Linux" optional component to use WSL1.
PS C:\Windows\system32> winget install -e --id Docker.DockerDesktop
The 'msstore' source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes [N] No: Y
Found Docker Desktop [Docker.DockerDesktop] Version 4.91.0
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://desktop.docker.com/win/main/amd64/239619/Docker%20Desktop%20Installer.exe
██████████ 598 MB / 598 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Windows\system32>
```

Or download the installer from <https://www.docker.com/products/docker-desktop/> and run it. Keep "Use WSL 2 instead of Hyper-V" ticked.

Restart if prompted, launch Docker Desktop, and accept the terms. Then verify:

```
LENOVO@LENOVO MINGW64 ~  
$ docker --version  
Docker version 29.8.0, build 88096ef
```

Docker Desktop has a built-in single-node Kubernetes cluster you can switch on from settings.

Enable Kubernetes in Docker Desktop

1. Open **Docker Desktop** and wait until the engine shows as running (green status at the bottom left).
2. Click the **gear icon** (Settings) at the top right.
3. Select **Kubernetes** in the left menu.
4. Toggle **Enable Kubernetes** on.
5. Choose a cluster type if your version offers it:
 - o **Kubeadm**: a single-node cluster. Simplest option and fine for most local work.
6. Click **Apply** (or **Apply & restart / Install**). Docker will download the Kubernetes images, which can take several minutes on the first run.
7. Wait until the Kubernetes status at the bottom left turns green.

```
LENOVO@LENOVO MINGW64 ~
$ kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1

LENOVO@LENOVO MINGW64 ~
$ kubectl config get-contexts
CURRENT   NAME             CLUSTER             AUTHINFO             NAMESPACE
*         docker-desktop   docker-desktop      docker-desktop
LENOVO@LENOVO MINGW64 ~
$ kubectl config use-context docker-desktop
Switched to context "docker-desktop".

LENOVO@LENOVO MINGW64 ~
$ kubectl get nodes
NAME             STATUS    ROLES    AGE     VERSION
docker-desktop   Ready     control-plane   3d21h   v1.36.1
```

Install Python 3.13.3

```
#winget install -e --id Python.Python.3.13 --version 3.13.3
```

```
LENOVO@LENOVO MINGW64 ~
$ python --version
Python 3.13.3

LENOVO@LENOVO MINGW64 ~
$ pip --version
pip 25.0.1 from C:\Users\LENOVO\AppData\Local\Programs\Python\Python313\Lib\site-packages\pip (python 3.13)
```

Sample Pod Deployment in k8 Cluster

Created namespace, a 3-replica Deployment, and a ClusterIP Service for nginx in the **web** namespace, with resource requests/limits and readiness/liveness probes set.

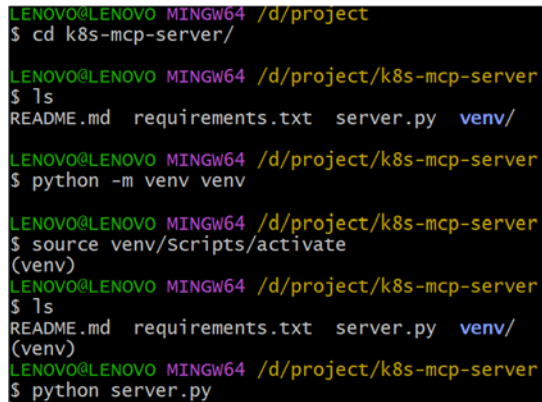
Apply with:

```
#git clone https://github.com/weldonfaizan/k8s-mcp-server.git
# cd k8s-mcp-server/deployment/nginx-deployment.yaml
#kubectl apply -f nginx-deployment.yaml
```

MCP Server Setup

Clone the repo in your laptop and prepare the environment to run MCP Server.

```
# git clone https://github.com/weldonfaizan/k8s-mcp-server.git
# cd k8s-mcp-server
# python -m venv venv
# source venv/bin/activate # on Windows: venv\Scripts\activate
# pip install -r requirements.txt
```

A terminal window with a black background and green text. The prompt is 'LENOVO@LENOVO MINGW64 /d/project'. The user enters '\$ cd k8s-mcp-server/'. The prompt changes to 'LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server'. The user enters '\$ ls', and the output is 'README.md requirements.txt server.py venv/'. The user enters '\$ python -m venv venv'. The prompt changes to 'LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server'. The user enters '\$ source venv/Scripts/activate', and the prompt changes to '(venv)'. The user enters '\$ ls', and the output is 'README.md requirements.txt server.py venv/'. The user enters '\$ python server.py'.

```
LENOVO@LENOVO MINGW64 /d/project
$ cd k8s-mcp-server/

LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server
$ ls
README.md requirements.txt server.py venv/

LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server
$ python -m venv venv

LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server
$ source venv/Scripts/activate
(venv)
LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server
$ ls
README.md requirements.txt server.py venv/
(venv)
LENOVO@LENOVO MINGW64 /d/project/k8s-mcp-server
$ python server.py
```

Run standalone (sanity check)

```
#python server.py
```

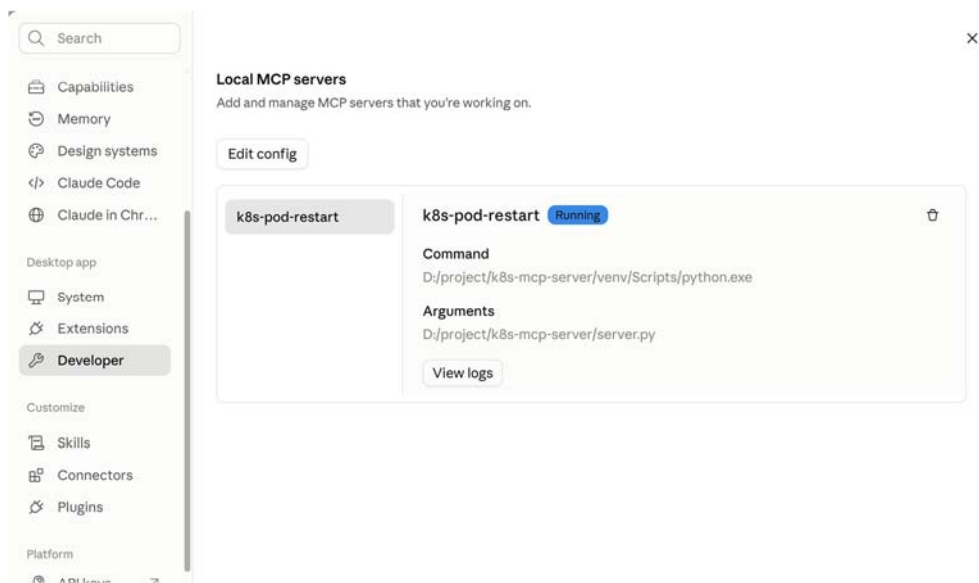
This starts the server over stdio and waits for an MCP client to connect — it won't print anything on its own, that's expected.

Wire it into an MCP client – Claude desktop

Add it to your client's MCP server config, e.g.:

```
{
  "mcpServers": {
    "k8s-pod-restart": {
      "command": "/absolute/path/to/venv/bin/python",
      "args": ["/absolute/path/to/k8s-mcp-server/server.py"]
    }
  }
}
```

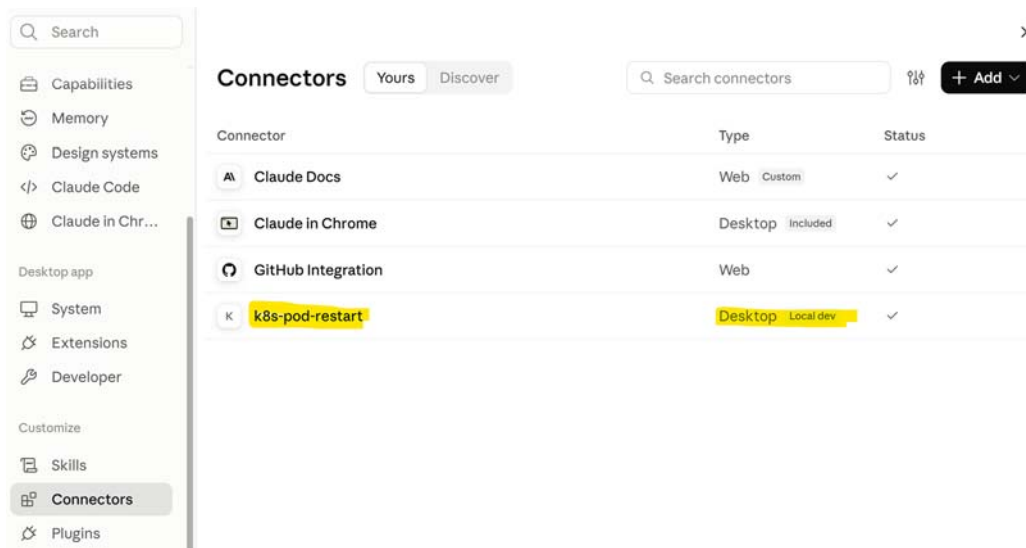
1. Open Claude Desktop > File > Settings > Desktop App: Developer



2. This either opens the existing file or creates a blank one at the correct location automatically.
3. Paste your JSON in (drop the jsonc comments-style syntax if any sneak in — it must be valid plain JSON):

```
{ } claude_desktop_config.json X
C: > Users > LENOVO > AppData > Local > Packages > Claude_pzs8sxrjxfjc > LocalCache > Roaming > Claude > { } claude_desktop_config.json >
1 {
2   "mcpServers": {
3     "k8s-pod-restart": {
4       "command": "D:/project/k8s-mcp-server/venv/Scripts/python.exe",
5       "args": [
6         "D:/project/k8s-mcp-server/server.py"
7       ]
8     }
9   },
}
```

4. Save, then fully quit Claude Desktop (on Windows, quit from the system tray — closing the window isn't enough) and reopen it.
5. Check the "+" → Connectors menu to confirm k8s-pod-restart shows up.



Testing Final Output with MCP Client:

Kubernetes pod management  

list pods in web namespace

K8s pod restart: List pods >

Pods in the `web` namespace:

- `nginx-deployment-65b7dcf6-jmnjn` (Running)
- `nginx-deployment-65b7dcf6-rrhdr` (Running)
- `nginx-deployment-65b7dcf6-w2t8b` (Running)

All three are up. Let me know if you want one restarted.

Kubernetes pod management  

restart this pods `nginx-deployment-65b7dcf6-jmnjn`

K8s pod restart: Restart pod >

Done — `nginx-deployment-65b7dcf6-jmnjn` was deleted. It's part of the `nginx-deployment` ReplicaSet, so Kubernetes should spin up a replacement automatically. Want me to check `list_pods` now to confirm the new one is up?

list pods in web namespace

K8s pod restart: List pods >

Confirmed — the replacement pod `nginx-deployment-65b7dcf6-lkxq` is up and Running, alongside the other two unaffected pods.

Reply 

Claude is AI and can make mistakes.

Sonnet 5 Medium Manual